Aniketh Tarikonda

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Education

University of Illinois, Urbana-Champaign Champaign, IL
Bachelor of Science in Computer Engineering, Minor in Physics Expected Dec. 2027

- Honors: James Scholar, Dean's List (Fall 2024–Present), Member of IEEE-HKN Alpha Chapter
- Cumulative GPA: 3.96 / 4.00
- Relevant Coursework: Computer Organization and Design (**Computer Architecture**, **Synopsys Verdi/VCS**), Intro to VLSI System Design (**Cadence Virtuoso/Innovus**), Applied Parallel Programming (**GPU Architecture & CUDA**), Digital Systems Laboratory (**FPGAs**), Digital Signal Processing, Analog Signal Processing, Data Structures

Experience

Eco Illini Supermileage Champaign, IL
Electrical/Firmware Lead Aug. 2024 – Present

- Leading an interdisciplinary team of 20+ engineers to design, verify, and integrate electrical systems for new generations of electric vehicles.
- Designed schematics and PCBs for 3+ custom boards, including a power distribution unit and battery management system, following automotive standards (e.g., SAE J1939).
- Programmed custom STM32 microcontroller-based boards with firmware supporting SPI, UART, and CAN communication protocols.
- Validated hardware on 2+ boards, achieving reliable data transmission rates exceeding 99%.
- Mentoring new members in automotive systems, embedded systems, and concepts in electrical engineering.

ResearchBase Inc. Pleasanton, CA
Software Engineering Intern July 2025 – Aug. 2025

- Built a ResearchCopilot prototype with a real-time conversational pipeline integrating automatic speech recognition (ASR) and analytics to surface insights dynamically.
- Improved document parsing accuracy by integrating LandingAI Agentic Document Extraction (ADE) APIs for structured data extraction.

Projects

Superscalar Out-of-Order RISC-V Processor Mar. 2026 - May 2026

- Designed a superscalar out-of-order processor implementing the RV32IM ISA, based on an explicit register renaming (ERR) architecture to enable high instruction-level parallelism.
- Developed core memory and speculative execution subsystems, including a non-blocking 4-way set-associative L1 data cache, 2-way L1 instruction cache, and gshare branch prediction.

FPGA Flight Simulator with 3D Rendering Nov. 2025 - Dec. 2025

- Designed and implemented a hardware-accelerated flight simulator on the RealDigital Urbana board using SystemVerilog, featuring a custom graphics pipeline, DDR3 SDRAM interfaces, and real-time 3D graphics rendering to a HDMI display.
- Developed a MicroBlaze-driven control subsystem to process keyboard inputs and transmit object data to the hardware GPU pipeline over GPIO interface(s).

GPT-2 Transformer Model Implementation Oct. 2025 - Nov. 2025

- Implemented core components of the GPT-2 transformer model in CUDA, and leveraged Nsight Compute for kernel-level profiling.
- Implemented optimizations to improve inferencing performance, including but not limited to KV-caching, reduction trees, and leveraging tensor-core-accelerated matrix operations.

Technical Skills

Languages: SystemVerilog, Verilog, C/C++, Python, CUDA, RISC-V, bash, tcl, Java, Javascript
Tools: Synopsys VCS, Verdi, Design Compiler, Cadence Virtuoso, Innovus, Calibre, Vivado, Vitis, Verilator, STM32CubeIDE, Linux, Altium Designer, KiCAD, Jira, oscilloscopes, function generators, gdb, git
Libraries/Frameworks: NumPy, Node.js, React, Spring, PostgreSQL